Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [18CS5IOT07]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

SOFTWARE ENGINEERING

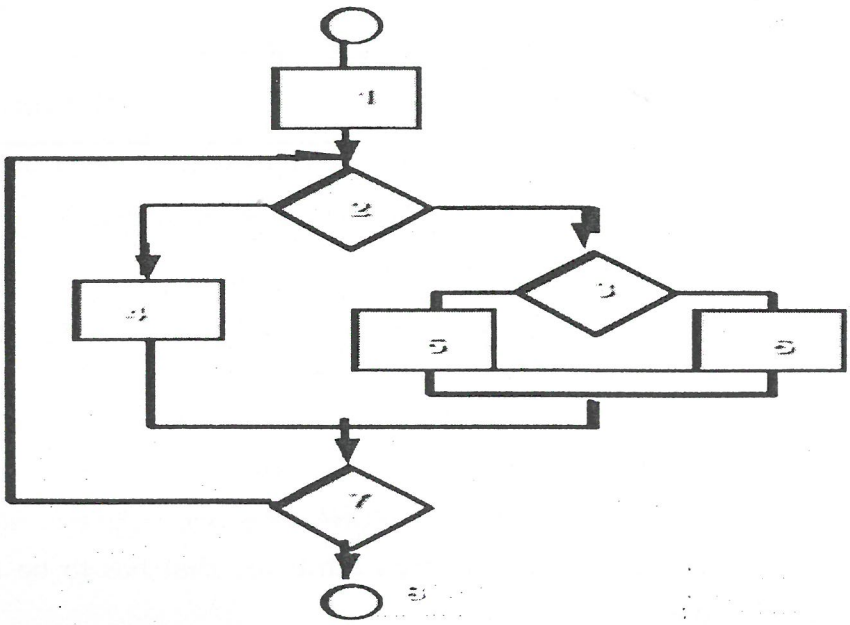
Time:3 Hours **Maximum Marks:70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	Let us assume that the DXC bank is planning to have a new banking ATM application to be developed and they have approached your company on 18.12..2021. Gather the Vijaya bank requirements and represent the development using the incremental process models.	**07**
	b.	Prof. Pavan Kalyan is developing a CORONA detecting software. Illustrate him with suitable examples about the good attributes that has to be incorporated with that software .	**07**
2.	**a.**	With a neat diagram, describe the different states of a Microwave oven using state machine model.	**07**
	b.	If ELGI is a MNC with a budget of 500 Crores, 5000 employees working in it, and using R1 technologies for its software production. A business proposal came with the requirements of 350 crores ,requiring 295 employees manpower for 3 months duration making use of RT technologies. Conduct the feasibility study on the new proposal and conclude the results.	**07**
3.	**a.**	Mr. Mickel is designing the user interface for a software, assist him to follow the Golden rules of user interface designing.	**07**
	b.	Industrialist Mr. Messi is deploying a data acquisition system for his agricultural field growing grapes. Lead him the way to carry out the deployment process providing him with the details of data acquisition system.	**07**
		OR	
4.	**a.**	If you are planning to design a software for JU, how can you make use of data-centered architecture software design style? Illustrate with a suitable example.	**07**
	b.	Revolutionary IT industrialist Mr. Ganguly, is venturing an advanced real time CORONA detection System . Appraise Mr. Ganguly the Real time software design paradigm along with efficient components of RTOS and its generic architecture.	**07**
5.	**a.**	GKL corporation, a Japanese based MNC owned by Mittel, is trying to optimize the NetSn software's overall performance. Discuss the usage of different integration testing methodologies in optimizing the overall performance of	**08**

NetSn software.

b.

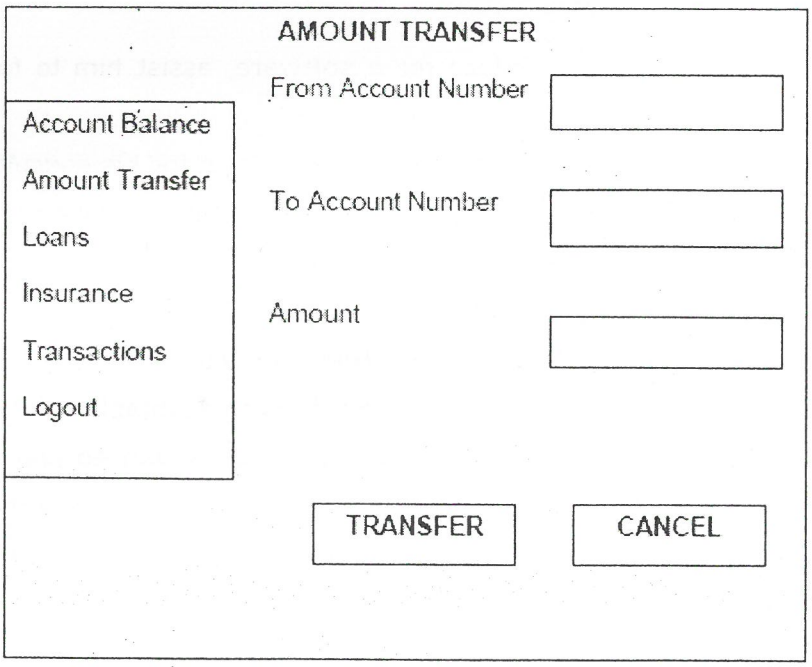


06

In the above figure, lead Mr. Ambrose is capturing the cyclomatic complexity concepts. Guide him in calculating the cyclomatic complexity and also to identify the possible paths for the above shown figure.

OR

6. a. Let us consider that Bank of Blitz wants a software for banking purpose and it asks the company KRSK Pvt.Ltd. to develop this software. When the user clicks his valid user name and enters his password, then he is taken into the homepage. The homepage is something as shown in below diagram :



07

Conduct the black box testing for amount transfer and account balance operations in the above diagram.

b. The following figure represents login page and registration page for MGM Pvt.Ltd. Verify these pages using black box testing and system testing. **07**

Assume -Password should should contain at least 8 characters, in which one

Capital letter, one special character, one numerical.

-Username minimum 8 characters, only alphabets and numerical are accepted

Login

Username or email address *

Password *

LOGIN Remember me

Lost your password?

Register

Email address *

Password *

REGISTER

7. a. Suppose that a project is estimated to be of 400 KLOC. Calculate the Effort,Team size,duration for the following categories of projects: **07**

	a	b	c	d
Organic	2.4	1.05	2.5	0.38
Semi-detached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32

b. Renowned industrialist Mr.Arav is venturing a software maintenance project. Elaborate on the following important aspects that he should be very well aware of his new venture: Types of software maintenance, Software Project planning **07**

OR

8. a. Let us consider that Bank of Bangalore wants a software for banking purpose and it asks the company ABD to develop the software. Educate ABD team members regarding the following: Principles of Risk management, Classification and categories of risks. **07**

b. Famous entrepreneur Mr.Nadal wants a forecasting process framework based on the results of multiple rounds of questionnaires sent to a panel of experts to take a business decision regarding his new venture. Identify and illustrate the method which Mr. Nadal can adopt in his decision making process. **07**